

A continuous squared-phase curve through the dimension-five SIC

FA/Banach run packet for arXiv:1903.06721

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Source conjecture and result

Appleby–Bengtsson–Flammia–Goyeneche conjecture in Section 6.5 of arXiv:1903.06721 that the squared-phase matrix for every SIC in dimension greater than 2 lies on a continuous curve. They construct curves in dimensions 3, 4, 6, 8, and report restricted defect 1 for the unique known dimension-five orbit 5a, but give no dimension-five curve.

This packet proves that missing case.

Theorem 1. *The squared-phase matrix associated with the exact Weyl–Heisenberg SIC on Scott–Grassl orbit 5a lies on a nonconstant real-analytic one-parameter curve of squared-phase matrices satisfying equations (52)–(54) of arXiv:1903.06721. Consequently, all six associated structures in that paper (the projector, Hermitian Hadamard matrix, two ETFs, and two symmetric tight fusion frames) also lie on continuous curves of structures of the same type.*

This is a partial result for the source’s all-SIC conjecture: it settles orbit 5a, not arbitrary dimensions or hypothetical SICs outside the known Weyl–Heisenberg orbit.

Weyl operators and the four-phase ansatz

Work over $\mathbb{Z}_5$, put $\tau = -e^{\pi i/5} = e^{6\pi i/5}$, and let

$$D_{(a,b)}e_j = \tau^{b(a+2j)}e_{j+a} \quad (a, b, j \in \mathbb{Z}_5).$$

For unit complex numbers u, v, w, z , define

$$M(u, v, w, z) = \begin{pmatrix} 1 & u & v & \bar{v} & \bar{u} \\ u & \bar{u} & w & z & w \\ v & \bar{w} & \bar{v} & z & z \\ \bar{v} & \bar{z} & \bar{z} & v & w \\ \bar{u} & \bar{w} & \bar{z} & \bar{w} & u \end{pmatrix}. \quad (1)$$

Thus $M_0 = 1$, $|M_p| = 1$, and $M_{-p} = \overline{M_p}$.

The exact orbit-5a fiducial in Scott–Grassl, arXiv:0910.5784, Appendix B has unnormalized vector ϕ expressed by nested square roots (the source-archive file is `sicsymbolic_5a.txt`). Set

$$M_p^\phi = \begin{cases} 1, & p = 0, \\ 6(\langle \phi, D_p^* \phi \rangle / \langle \phi, \phi \rangle)^2, & p \neq 0. \end{cases}$$

An exact finite Clifford/displacement normalization is

$$\widetilde{M}_{(x,y)} = \tau^{-4(4y-x)} M_{(3x+y, 2x+y)}^\phi. \quad (2)$$

Direct substitution of the published radicals gives $\widetilde{M} = M(u_0, v_0, w_0, z_0)$. Numerically (only for readability),

$$\begin{aligned} u_0 &= -0.073937094602198 + 0.997262907182347i, \\ v_0 &= 0.877119421692956 - 0.480272339500221i, \\ w_0 &= -0.272762046860239 + 0.962081527622588i, \\ z_0 &= -0.860547299152712 - 0.509370538921297i. \end{aligned}$$

Equation (2) preserves the squared-phase equations: its linear part has determinant one modulo 5, and its prefactor is a character. The finite radical substitution and the pattern identities are reproduced in the verifier accompanying this packet.

The 1 + 2 + 2 symmetry reduction

Let $F_0 = \text{diag}(1, 3)$ over $\mathbb{Z}_5$ and define

$$A(u, v, w, z) = \frac{1}{5} \sum_{p \in \mathbb{Z}_5^2} M(u, v, w, z)_{F_0 p} D_p. \quad (3)$$

This is Hermitian. Orthogonality of the Weyl operators and unimodularity of the coefficients give, identically throughout the four-torus,

$$\text{Tr } A = 1, \quad \text{Tr}(A^2) = 5. \quad (4)$$

The pattern (1) is invariant under $T = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. Consequently A commutes with the order-three unitary

$$U_{jk} = \frac{1}{\sqrt{5}} \tau^{2j^2 + 4jk}, \quad U D_p U^* = D_{Sp}, \quad S = F_0^{-1} T F_0 = \begin{pmatrix} 0 & 2 \\ 2 & 4 \end{pmatrix}. \quad (5)$$

A direct finite geometric-sum calculation shows that the three eigenspaces of U have dimensions 1, 2, 2. Put $\omega = e^{2\pi i/3}$ and let

$$P_0 = \frac{I + U + U^2}{3}, \quad P_1 = \frac{I + \omega^2 U + \omega U^2}{3};$$

the omitted spectral projector P_2 also has rank 2.

Write $u = e^{ia}, v = e^{ib}, w = e^{ic}, z = e^{id}$, and define the real-analytic map $f : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ by

$$f(a, b, c, d) = (\text{Tr}(P_0 A) - 1, \text{Tr}(P_1 A), \text{Tr}(P_1 A^2) - 2). \quad (6)$$

Lemma 1. *If $f(a, b, c, d) = 0$, then $A(a, b, c, d)^2 = I_5$.*

Proof. Since A commutes with U , it is a direct sum of Hermitian blocks A_0, A_1, A_2 of sizes 1, 2, 2. The first equation gives $A_0 = (1)$. The next two give $\text{Tr } A_1 = 0$ and $\text{Tr}(A_1^2) = 2$, hence $A_1^2 = I_2$. The automatic identities (4) now imply $\text{Tr } A_2 = 0$ and $\text{Tr}(A_2^2) = 2$, hence $A_2^2 = I_2$ as well. (For a Hermitian 2×2 matrix, trace zero and squared trace two force its eigenvalues to be 1, -1.) $\square$

A rigorous Jacobian certificate

At the exact radical point $x_0 = (\arg u_0, \arg v_0, \arg w_0, \arg z_0)$, the SIC equations and the equivalence lemma in Section 6 of arXiv:1903.06721 give $A(x_0)^2 = I_5$, so $f(x_0) = 0$. Differentiating (6), an outward-rounded interval calculation from the exact radicals gives

$$Df(x_0) \approx \begin{pmatrix} -0.645442112 & 0.310838788 & 0.237839634 & -0.125923322 \\ -0.083393887 & -0.351000491 & -0.752846190 & -0.272668334 \\ 1.302407962 & 0.889323925 & -0.018286153 & -1.338447426 \end{pmatrix}.$$

More importantly, the verifier certifies

$$\det D_{(a,c,d)}f(x_0) \in [-0.881829962934411, -0.881829962934410]. \quad (7)$$

All operations in this certificate are rational arithmetic, principal square roots, and outward-rounded interval operations; the code contains the full Scott–Grassl radical vector rather than the displayed decimals.

It follows from (7) and the real-analytic implicit-function theorem that, for b in an interval about b_0 , there are real-analytic functions $a(b), c(b), d(b)$ with $f(a(b), b, c(b), d(b)) = 0$. Because b itself varies, the phase matrix (1) is nonconstant. Lemma 1 gives $A^2 = I_5$ along the curve. The equivalence lemma of the source paper then converts this into the squared-phase equations and all six associated families. This proves Theorem 1.

Verification, novelty, and limitations

The script `code/verify_interval.py` prints the four exact-base enclosures, all twelve Jacobian-entry enclosures, and (7). It runs with the repository’s sandbox Python environment and uses `mpmath.iv`. The exact normalization and symmetry matrices can also be checked by finite sums over $\mathbb{Z}_5$.

The run indexes and a bounded arXiv/web search on 13 August 2026 were checked using the exact conjecture phrase and close orbit-5a variants. The search found the source and restricted-defect discussions, but no explicit curve or proof for orbit 5a. This is a likely-valid new partial result, not a claimed full solution of the all-dimensional conjecture.

A deeper upgrade attempt considered arbitrary Zauner-symmetric SICs. Here all symmetry blocks have size at most two. Larger blocks need higher spectral conditions, but at an involution the differentials of $\text{Tr}(B^k)$ collapse to combinations of $\text{Tr}(dB)$ and $\text{Tr}(B dB)$. Thus the larger-dimensional system is singular and needs a genuinely second-order or explicit idea.

References

- M. Appleby, I. Bengtsson, S. Flammia, and D. Goyeneche, *Tight Frames, Hadamard Matrices and Zauner’s Conjecture*, arXiv:1903.06721.
- A. J. Scott and M. Grassl, *SIC-POVMs: A new computer study*, J. Math. Phys. 51 (2010), 042203; arXiv:0910.5784.

Human-review recommendation. Review as a likely valid substantial partial result, prioritizing the finite normalization (2), the 1+2+2 block argument, and the interval determinant (7).